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Assignment 2: Electronic Door Lock on FPGA (Individual work)

Aim: FPGA programming and interfacing.

Design and implement a digital circuit that receives input from a keypad and operates in two distinct modes: **Locked** and **Unlocked**. The current mode should be indicated using an LED.

When the system is in the **Locked** mode, the digits entered through the keypad must be interpreted as an unlocking code. Once the correct code has been entered, the system transitions into the **Unlocked** mode. All entered digits should be displayed on a **7-segment display**.

When the system is in the **Unlocked** mode, pressing the **✱ (star)** key or **waiting 30 seconds** should return the circuit to the **Locked** mode.

Hint: To ensure stable operation with the keypad, consider using a low clock frequency (below 10 kHz).

You must simulate the complete circuit using pad_mod_behav.vhd.
After a successful simulation, implement the design on the FPGA, connect the provided keypad, and verify that the 7-segment display and LED respond correctly.

Bonus (Optional):

Improve the energy efficiency of the keypad encoder so that the internal counter does not run continuously when no key is pressed.
Example reference: https://www.youtube.com/watch?v=h-NM1xSSzHQ

Documentation Requirements

Your report should include:

- **Description of the components used**
- **Explanation of the circuit's functionality**
- **Validation strategy and results**

Use relevant **code excerpts**, **simulator screenshots**, and **photos of the FPGA setup** to illustrate your work. Additionally, create a **YouTube video** demonstrating the system in action.

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Submission Details

- **Submission platform:** Itslearning → Assignments
- **File format:** PDF
- **Deadline: 20/04/2026, before 23:59**
- **Page limit:** Maximum **3 pages**
- **Minimum font size: 10 pt**